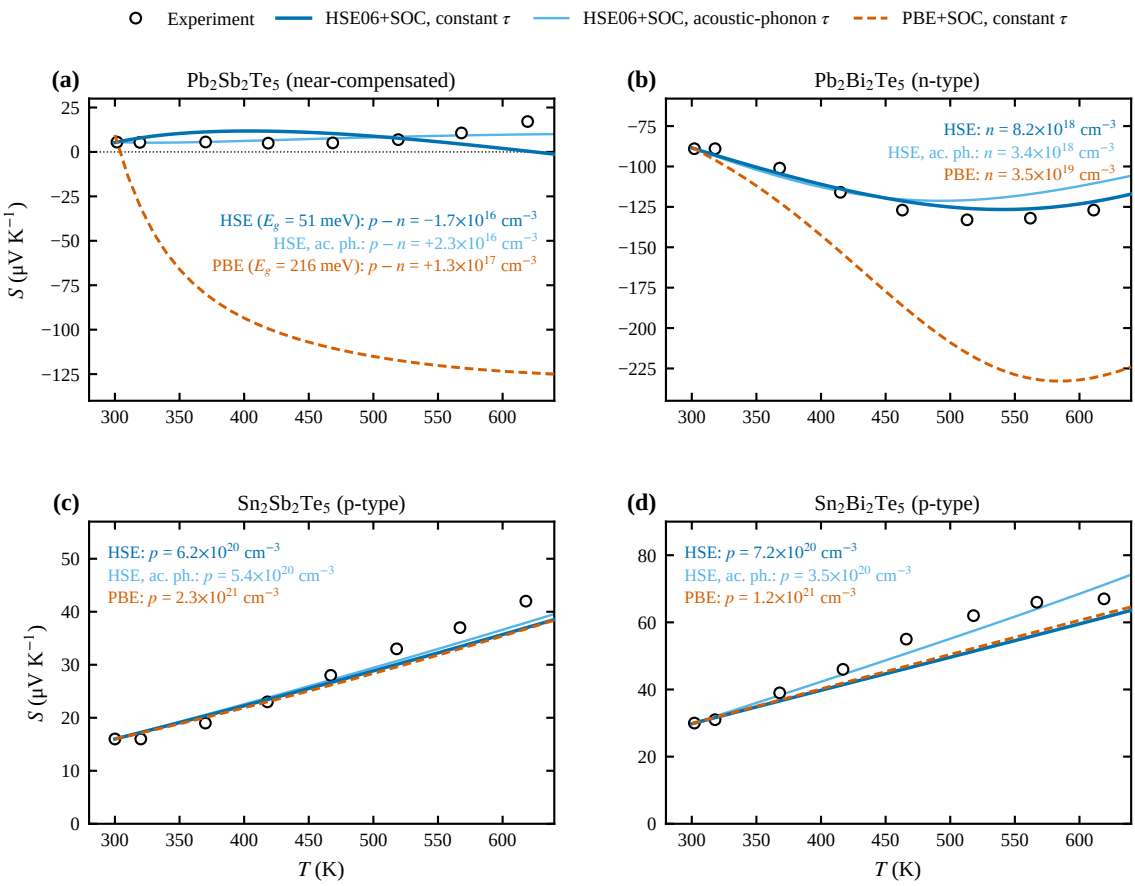
Experiment
HSE06+SOC, constant τ
HSE06+SOC, acoustic-phonon τ
PBE+SOC, constant τ
(a)
Pb2Sb2Te5 (near-compensated)
HSE (Eg = 51 meV): p − n = −1.7×10^16 cm^−3
HSE, ac. ph.: p − n = +2.3×10^16 cm^−3
PBE (Eg = 216 meV): p − n = +1.3×10^17 cm^−3
(b)
Pb2Bi2Te5 (n-type)
HSE: n = 8.2×10^18 cm^−3
HSE, ac. ph.: n = 3.4×10^18 cm^−3
PBE: n = 3.5×10^19 cm^−3
(c)
Sn2Sb2Te5 (p-type)
HSE: p = 6.2×10^20 cm^−3
HSE, ac. ph.: p = 5.4×10^20 cm^−3
PBE: p = 2.3×10^21 cm^−3
(d)
Sn2Bi2Te5 (p-type)
HSE: p = 7.2×10^20 cm^−3
HSE, ac. ph.: p = 3.5×10^20 cm^−3
PBE: p = 1.2×10^21 cm^−3
S (μV K^−1)
T (K)